8051 Microcontroller Basics and Assembly language Programming

Introduction to Embedded System

- A system is a set of interrelated components which are designed/developed to perform a common tasks or to do some specific work . Embedded means something which is integrated or attached with another thing. An Embedded System is an system specially designed to perform a designated function with the help of its hardware and embedded software. These embedded systems can work without human intervention or with a little human intervention. Examples of embedded system are Fire alarm, Washing machine, Digital cameras, refrigerators, calculators, Home security system.

There are three components of embedded system

- Hardware
- Software
- Real time operating system

Hardware components are

Power supply : For the embedded system the power supply is the key component to provide the power to the embedded system circuit. supply of 5 V DC or below it can be given using battery or circuit.

Processor: Microprocessor or microcontroller can be used. Select processor 8 bit,16 bit,32 bit depending on application

Memory: There are two types of memories including ROM (read-only-memory) and RAM (random access memory). The former is called the code memory that stores the program code and is non-volatile which means it stays stored in the system when the power supply is removed. While latter is called the data memory and is a volatile memory which means it is used for temporally storing the information and is removed from the system when the power supply is turned off.

Timers counters: Timers are used to create delays. While at times you want to count the number of times a particular event occurs. Counters are used in

such cases. If an up counter is used in the system, it will count up from the initial value to 0xFF and if it is down counter, it will count down to 0x00. The counters are integrated using register-type circuits like a flip-flop.

Communication ports : Communication ports are used in embedded systems to establish communication with other embedded systems. There are several communication ports including USB, UART, USB, I2C, SPI, and RS-485

Output and Input : Input is required to interact with the embedded system. A sensor can be used to provide input to the system. The microcontroller used in the system can be configured as an input or output port. In the microcontroller, there are a fixed number of input and output ports that you can utilize as per your requirement.

Circuits used in application : application specific circuits uses DMA,A/D,PWM for D/A and MODEM

Software components are software tools for programming and coding .They are as follows

- **Assembler:** The assembler tool converts the written code into a machine language. The assembler first converts source code to object code and then to the language that the machine can understand.

- **Emulator** : The main task of the emulator is to make the embedded system act like a real system in a simulation environment. It is used to simulate software performance, and it helps in achieving the ideal performance of the written code. With an emulator, you can run one operating system into another device. For example, using an emulator you can run Mac operating system into your windows operating system.

- **Debugger**: A debugger is a tool used for testing and debugging purposes. It scans the code thoroughly and removes the errors and bugs, and identifies the places where they occur.

Compiler: A **compiler** is used to turn this written code into low-level machine language that the machine can comprehend. The main purpose of this tool is to develop an executable program.

RTOS(Real time operating system): This supervises application software and provide mechanism to let processor run process as per scheduling. It defines way system works A small scale embedded system may not have RTOS.

Characteristics of embedded system:

-Require less power

- Low cost
- Task specific
- Time specific
- High efficiency, highly stable and Reliable
- Minimal user interface

Types of Embedded System:

- Classification based on two factors
  - Performance and Functional Requirements
  - Performance of Micro-controllers
- **Based on performance and functional requirements it is divided into 4 types**

1)Real-Time Embedded Systems

2) Stand Alone Embedded Systems

3) Networked Embedded Systems :

4) Mobile Embedded Systems **:**

- **Based on performance of Microcontrollers it is classified as**

1)Small Scale Embedded Systems

2)Medium Scale Embedded Systems

3)Sophisticated or Complex Embedded Systems

Applications:

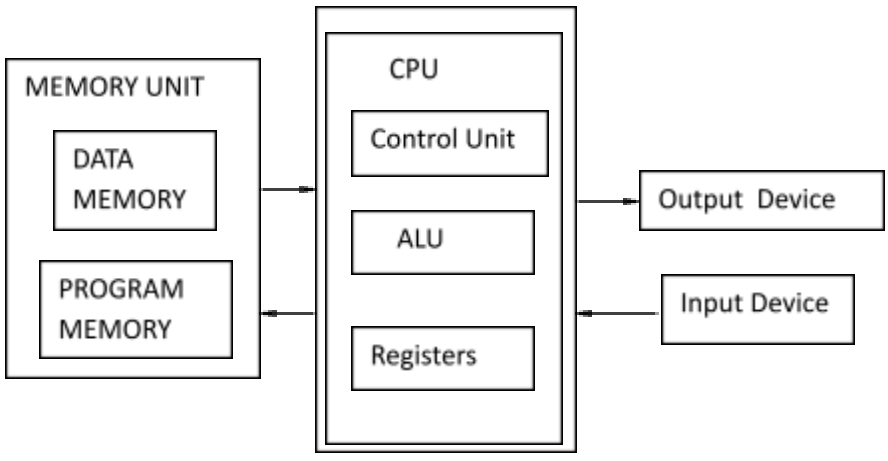
- Home appliances
- Transportation
- Health care
- Business sector & offices
- Defense sector
- Aerospace
- Agricultural Sector

Types of Architectures

Computer architecture is a set of rules and methods that describe the functionality, organization, and implementation of computer systems. There are two types of architectures .These are Von-Neumann architecture and Harvard architecture.

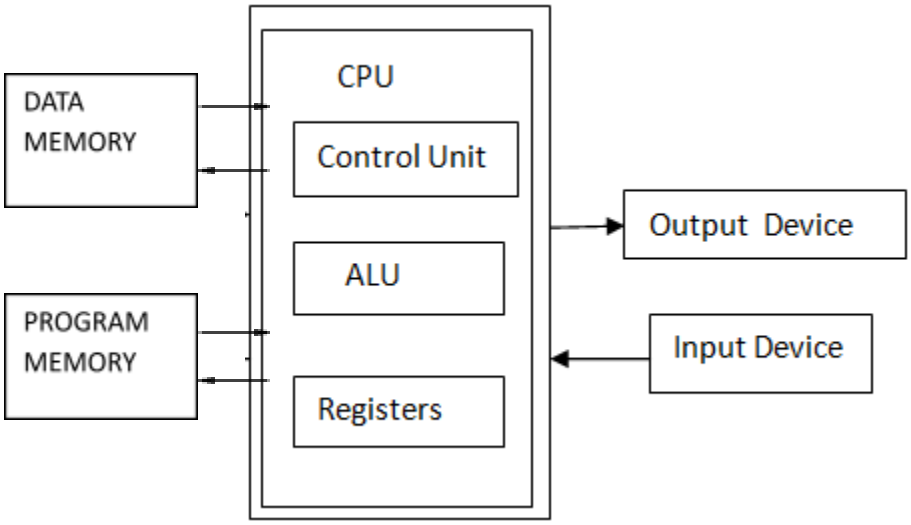
VON -NEUMANN Architecture

Von Neumann Architecture is a digital computer architecture whose design is based on the concept of stored program computers where program data and instruction data are stored in the same memory. This architecture was designed by the famous mathematician and physicist **John Von Neumann** in 1945.



Harvard Architecture

Harvard Architecture is the computer architecture that contains separate storage and separate buses (signal path) for instruction and data. It was basically developed to overcome the bottleneck of Von Neumann Architecture. The main advantage of having separate buses for instruction and data is that the CPU can access instructions and read/write data at the same time.



Difference between Von-Neumann and Harvard Architecture

VON-NEUMANN	HARVARD
1)It is ancient computer architecture based on stored program computer concept.	1) It is modern computer architecture based on Harvard Mark I relay based model.
2)Same physical memory address is used for instructions and data.	2)Separate physical memory address is used for instructions and data.
3)There is common bus for data and instruction transfer.	3)Separate buses are used for transferring data and instruction.
4)Two clock cycles are required to execute single instruction.	4)An instruction is executed in a single cycle.

VON-NEUMANN	HARVARD
6) CPU can not access instructions and read/write at the same time.	CPU can access instructions and read/write at the same time.
7)It is used in personal computers and small computers.	It is used in micro controllers and signal processing.

RISC VS CISC

Reduced Instruction Set Architecture (RISC) –

The main idea behind this is to make hardware simpler by using an instruction set composed of a few basic steps for loading, evaluating, and storing operations just like a load command will load data, a store command will store the data. A microprocessor architecture with a simple collection and highly customized set of instructions . It is built to minimize the instruction execution time by optimizing and limiting the number of instructions.

Features of RISC Processor

1. **One cycle execution time:** For executing each instruction in a computer, the RISC processors require one CPI (Clock per cycle). And each CPI includes the fetch, decode and execute method applied in computer instruction.
2. **Pipelining technique:** The pipelining technique is used in the RISC processors to execute multiple parts or stages of instructions to perform more efficiently.
3. **A large number of registers:** RISC processors are optimized with multiple registers that can be used to store instruction and quickly respond to the computer and minimize interaction with computer memory.
4. It supports a simple addressing mode and fixed length of instruction for executing the pipeline.

5. It uses LOAD and STORE instruction to access the memory location.
6. Simple and limited instruction reduces the execution time of a process in a RISC.

CISC(Complex Instruction Set Architecture)

The main idea is that a single instruction will do all loading, evaluating, and storing operations just like a multiplication command will do stuff like loading data, evaluating, and storing it, hence it's complex. CISC chips are relatively slower as compared to RISC chips but use little instruction than RISC. It has a large collection of complex instructions that range from simple to very complex .

CISC Features:

1. Complex instruction, hence complex instruction decoding.
2. Instructions are larger than one-word size.
3. Instruction may take more than a single clock cycle to get executed.
4. Less number of general-purpose registers as operations get performed in memory itself.
5. Complex Addressing Modes.
6. More Data types.

Example – Suppose we have to add two 8-bit numbers:

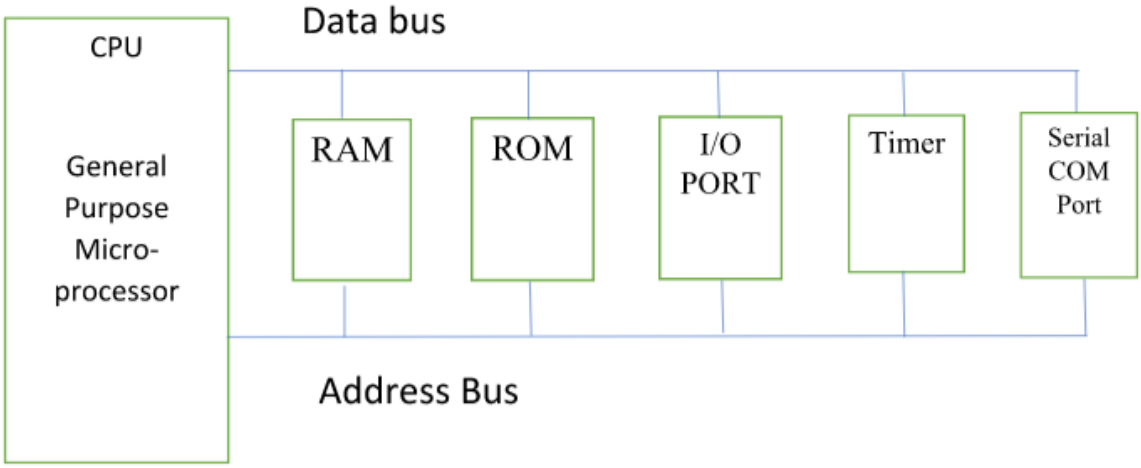
- **CISC approach:** There will be a single command or instruction for this like ADD which will perform the task.

- **RISC approach:** Here programmer will write the first load command to load data in registers then it will use a suitable operator and then it will store the result in the desired location.

MICROPROCESSOR and MICROCONTROLLER

Microprocessor

A microprocessor is an integrated circuit that consist only CPU (Central Processing Unit). It consists of an ALU, control unit and many registers, Where ALU performs arithmetic and logical operations on the data received from an input device or memory. Control unit controls the instructions and flow of data within the computer and, number of registers like Register B, Register C, Register D, Register E, Register H, Register L, and accumulator. Operations performed by microprocessor are adding, subtracting, comparing two numbers and fetching the data from memory for transferring it from one place to another.



Let's see the basic components of microprocessor:

Arithmetic and Logic unit (ALU): ALU inside a microprocessor used to perform the arithmetic and logic operation. It performs the logic operation on the data stored inside a register.

Accumulator: It is a register inside which the intermediate arithmetic and logical operation data is stored.

Working registers: It is a storage device used to store the data inside a microprocessor in different address location.

Program counter: It is used for counting the number of program executed inside a microprocessor.

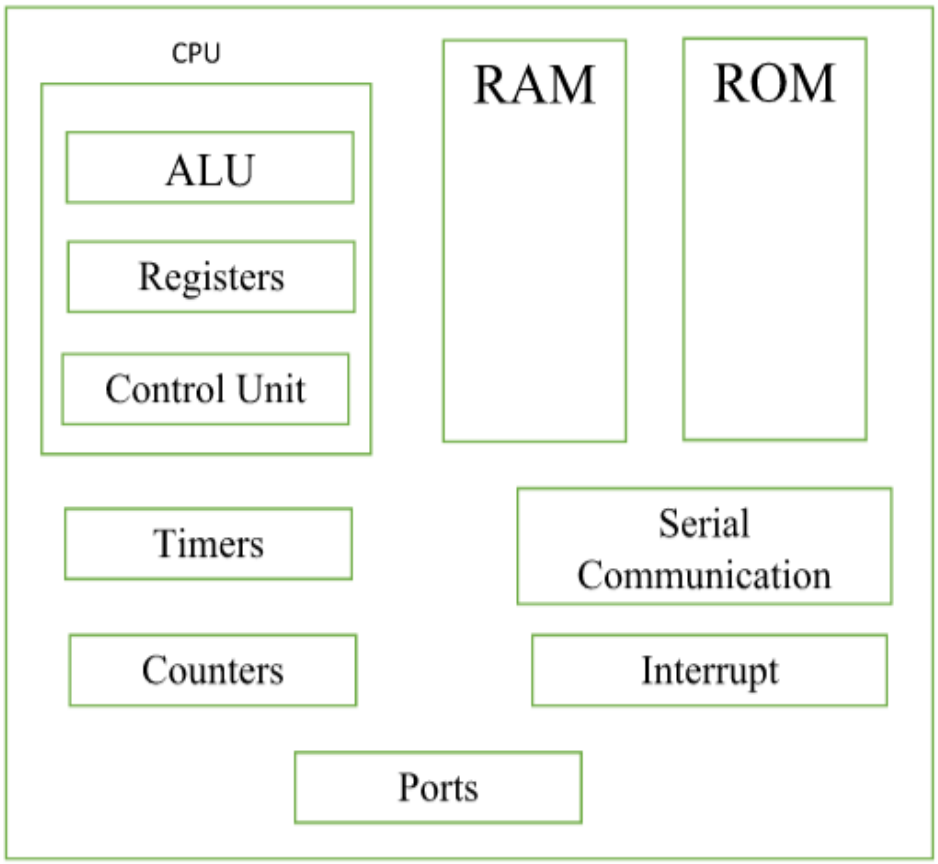
Stack pointer: Stack pointer act as a pointer to the certain address. It is a register used to store the address of the last program request made by the processor inside a stack.

Clock circuit: It is used for generate the clock pulse required as a reference signal for the microprocessor.

Microcontroller

Microcontroller is a microprocessor based integrated peripherals. It contain CPU, ROM, RAM, timers, counters, I/O ports, ADC, DAC, serial communication all functional block on a single chip, which make it a complete system. It is dedicated to performing a particular task and execute one specific application. We cannot modify the size of RAM, ROM and other components. Once a controller is designed the structure is fixed. So, the structure of the microcontroller is not flexible.

Block diagram of microcontroller is as shown below



Difference between microprocessor and Microcontroller

Microprocessor	Microcontroller
▪ Microprocessor is the heart of Computer system.	Micro Controller is the heart of an embedded system.
▪ It is only a stand alone processor, so memory and I/O components need to be connected externally.	Micro Controller has a processor along with internal memory and I/O components.

▪ Memory and I/O has to be connected externally, so the circuit becomes large.	Memory and I/O are already present, and the internal circuit is small.
▪ Power consumption high	Power consumption low
▪ Cost of the entire system is high.	Cost of the entire system is low.
▪ It uses an external bus to interface to RAM, ROM, and other peripherals.	It uses an internal controlling bus.
▪ Most of the microprocessors do not have power saving features.	Most of the microcontrollers offer power-saving mode.
▪ It is mainly used in personal computers.	It is used mainly in a washing machine, MP3 players, and embedded systems.
▪ It's used for general purpose applications that allow you to handle loads of data.	It's used for application-specific systems.
▪ Microprocessors are based on Von Neumann model.	Micro controllers arc based on Harvard architecture.